MAOPhot 1.2.2

Welcome to MAOPhot 1.2.1, a PSF photometry tool built with Astropy 7.1.1 and Photutils 2.3.0.

Version 1.2.1 Changes

1) Added much needed check for saturated (above linearity limit) target, check, and comp stars when calling Photometry→Get Comparison Stars

Version 1.2.1 Changes

2) Fixed problem when reloading photometry data

Version 1.2.0 Changes

1) **New feature:** Multi-color photometry

   Adds support for 3-color and 4-color transformed reports. Analysis uses the AAVSO-recommended iterative solution, following the approach used by AAVSO TransformApplier (v2.7.1).

   MAOPhot applies the AAVSO-recommended transformation coefficients.

   MAOPhot requires the following coefficients:

coeff	B	V	R	I
BVRI	Tb_bv	Tv_bv	Tr_vi	Ti_vi
BVR	Tb_bv	Tv_bv	Tvr	
BVI	Tb_bv	Tv_bv		Ti_vi
VRI		Tv_vi	Tr_vi	Tvi
BV	Tb_bv	Tv_bv		
VR		Tv_vr	Tr_vr	
VI		Tv_vi		Tvi

2) Upgraded packages:
   a. Astropy 7.2.0
   b. NumPy 2.4.0
   c. Pandas 3.0.1
   d. Pillow 12.1.1
   e. SciPy 1.17.1

f. tqdm 4.67.3

3) Settings changes:
    a. Added *Oversampling*, used when building the ePSF model (default: 2).
    b. Added additional transformation coefficients (e.g., *Tb_br*, *Tb_bi*, *Tr_ri*).
    c. Added *User Note*.
    d. Added *Date-Obs (UTC)*.
    e. *Date-Obs (JD)* and *Date-Obs (UTC)* are read-only and are obtained from the FITS file.
4) Saved the current oversampling value in the exported ePSF file.
5) Corrected the single-comparison-star error calculation (uses 1/SNR).
6) Added *Minimum SNR* to Settings. If any comparison star or check star has an SNR below the minimum, that star is excluded.
7) Corrected a case where the check star was not labeled.
8) Reports now record the correct transformation coefficients in the Notes section of the AEFF (AAVSO Extended File Format) report.
9) AAVSO report names are now derived from the Master Report name rather than the loaded FITS file.

Introduction

Aperture photometry and point spread function (PSF) photometry are two widely used methods for measuring stellar brightness (magnitude). The sections below summarize how they differ.

1. Aperture Photometry

a. Definition:

- Aperture photometry involves summing the light within a circular (or elliptical) aperture centered on the star and subtracting the background light estimated from an annulus around the aperture.

b. Advantages:

- Simple and straightforward to implement
- Works well for isolated stars with little contamination from nearby stars or artifacts

c. Disadvantages:

- Less effective in crowded fields, where light from nearby stars may contaminate the aperture
- Accuracy depends on the choice of aperture size; a large aperture captures more light but also more background noise, while a small aperture might miss some of the star's light
- When stars are blended, it is obviously impossible simply to sum the CCD data numbers corresponding to the image of each star and then to subtract an allowance for the diffuse sky emission (quote from Stetson 1986)

d. Usage:

- Suitable for relatively uncrowded fields and for stars with a relatively high signal-to-noise ratio (SNR)

2. PSF Photometry

a. Definition:

- PSF photometry models the star's light distribution as a mathematical function called the Point Spread Function (PSF), which describes how the star's light spreads across the detector. The PSF is then fitted to the star to determine its total flux, accounting for overlaps with nearby stars.
- PSF photometry fits a model in which two or more expected stellar profiles are superimposed: each model profile is shifted in x and y and scaled in intensity, and one or more parameters describing the local distribution of diffuse sky light are varied until a satisfactory fit of the overall model to the image data is achieved (quote from Stetson 1986).

b. Advantages:

- Excellent for crowded fields, as it can deconvolve the light from overlapping stars
- Provides more precise measurements, especially in dense star clusters or regions with significant background contamination
- The PSF model can be a simple analytic function (e.g., 2D Gaussian or Moffat profile) or a more complex model derived from a 2D PSF image

c. Disadvantages:

- More computationally intensive and requires accurate modeling of the PSF

- Sensitive to systematic errors if the PSF model does not accurately represent the actual star profiles

d. Usage:

- Ideal for dense stellar fields, such as globular clusters or the cores of galaxies

About MAOPhot

This program is derived from “MetroPSF” by Maxym Usatov. It has been renamed and extended to generate AAVSO reports and to streamline creation of an effective PSF (ePSF) for PSF photometry.

MAOPhot calculates stellar magnitudes from FITS-formatted images using PSF photometry. It produces an AAVSO Extended File Format (AEFF) report that can be submitted to the AAVSO via WebObs.

MAOPhot uses PSF (point spread function) photometry exclusively.

MAOPhot is written in Python using Astropy (a common core package for astronomy) and Photutils. For background on the Photutils classes and methods used by MAOPhot, see the Photutils “PSF Photometry” user guide.

Key Features

MAOPhot is designed for AAVSO reporting and includes the following features:

- Built with Astropy and Photutils
- Generates an effective PSF (ePSF) model following Anderson and King (2000; PASP 112, 1360), with optional creation of a rejection list of stars to exclude from the ePSF build
- Option to use a Circular Gaussian PRF (Pixel Response Function) as a model
- Option to use a Moffat PSF model
- Supports **iterative PSF photometry** (Photutils IterativePSFPhotometry), where new sources are detected in the residual image after fitted sources are subtracted—useful for crowded fields
- PSF Photometry using an ensemble of comparison stars or a single comp star
- Generates multi-color photometry reports (BV, VR, VI, BVR, BVI, VRI, and BVRI) using transformation coefficients (TCs) in AAVSO extended format; single-image photometry reports are also supported
- Use of telescope Transformation Coefficients (needed for Color Photometry)
- User can specify check star and list of comp stars
- Manually select a star for measurement
- Intermediate results are saved as .csv files

- Optionally enter an AAVSO Chart ID when retrieving comparison star data
- When image file is loaded, optionally elect to find weighted median Moffat β value and weighted median FWHM value

Photutils Integration

MAOPhot leverages several key Photutils classes:

- **PSFPhotometry**: Provides a flexible framework for PSF photometry workflow that can find sources, group overlapping sources, fit the PSF model, and subtract fitted PSF models from the image
- **IterativePSFPhotometry**: Iteratively calls PSFPhotometry to detect new sources in the residual image after subtracting fitted sources - essential for crowded field photometry
- **DAOStarFinder**: Implements the DAOFIND algorithm (Stetson 1987) to detect stars by searching for local density maxima. The Gaussian kernel is defined by FWHM, ratio, theta, and sigma_radius parameters. DAOStarFinder finds object centroids by fitting the marginal x and y 1D distributions of the Gaussian kernel to the marginal x and y distributions of the input data image
- **LocalBackground & MMMBackground**: Used for local background estimation around each source before PSF fitting

System Requirements

1. MAOPhot runs on Windows 10 or Windows 11 (a Unix installation is also available).
2. 8 GB of memory and 1 GB storage

Installation

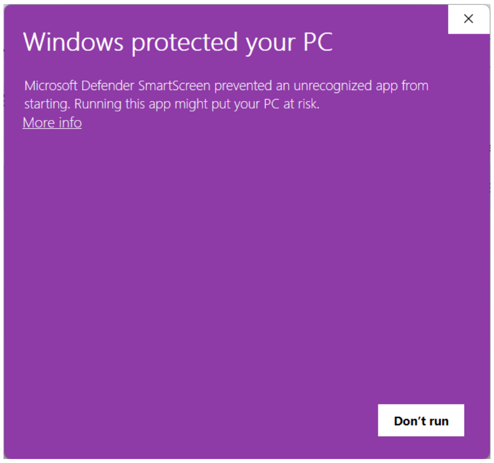
1. Download the installer from the MAOPhot releases page on GitHub.

   a. Under MAOPhot 1.2.0, scroll to *Assets*.

   b. Download *MAOPhot_SETUP.exe* (override any download warnings, if prompted).

2. Run *MAOPhot_SETUP.exe* on your computer.

a. You may see a Windows security warning (Windows 11).



b. Select *More info*, then *Run anyway*.

c. If prompted, allow the app to make changes to your device.

d. Accept the license agreement (MIT License, used by many Python applications).

e. Choose a destination directory.

f. Continue to install MAOPhot.

3. Launch *MAOPhot.exe*. Two windows should appear:
    - **Settings window**: Contains key parameters for PSF photometry, telescope specifications, and the target, comparison, and check star settings.
    - **Main window**: Contains image-stretching tools on the left, the image display and console in the center, and the PSF analysis section on the right

Single Image Photometry Workflow

General workflow for single-image photometry and AAVSO report generation, using the example "2022 8 1 V1117 Her (Example)".

1) Prepare master images

The master image should be calibrated and saved in valid FITS format. Crop the image so that no "black" (zero-ADU) border remains at the edges. The image does not need to be plate-solved, but RA and Dec values must be present for successful plate solving in MAOPhot (see Astrometry.net server settings, below). The example FITS files *V1117Her_B.fit* and *V1117Her_V.fit* are already cropped and plate-solved.

2) Launch MAOPhot

Run the following command in the working directory:

MAOPhot.exe

3) Fill in recommended settings values

a. Select *Load…* to load an existing settings file (for example: 2022 8 1 V1117 Her (Example)\V1117 Her Settings.txt).

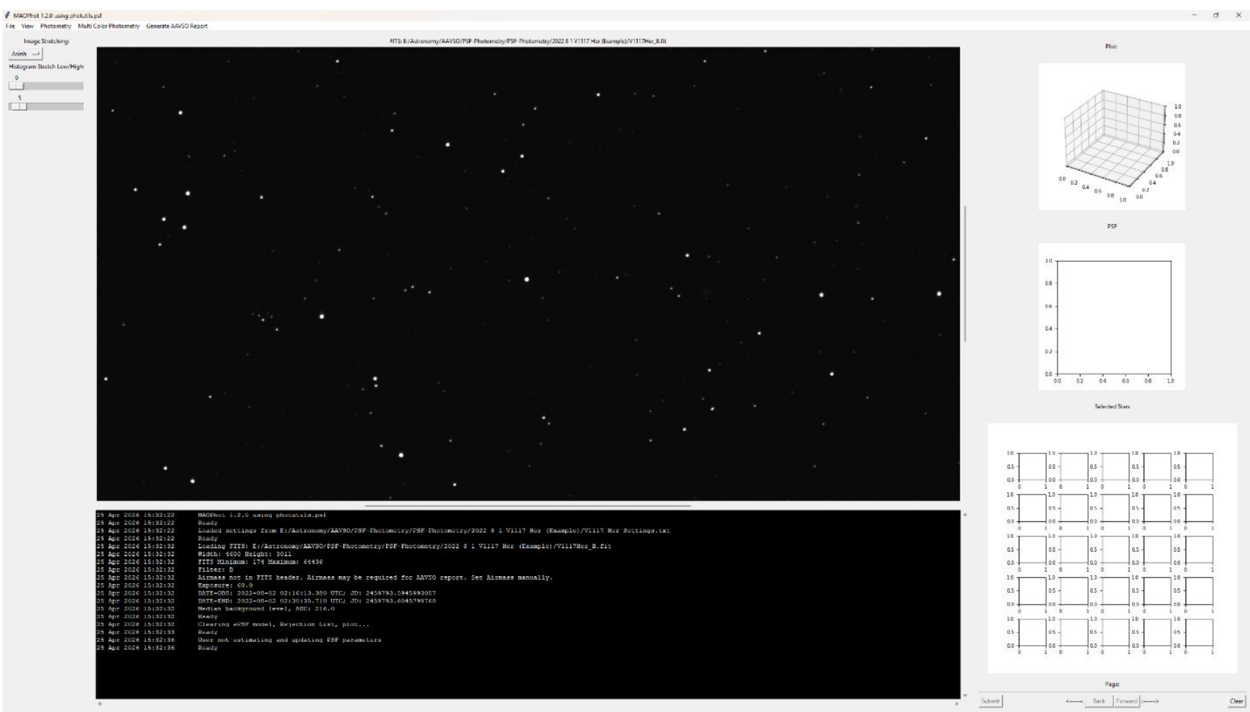
b. Close the Settings window by clicking *OK*.

4) Open the FITS file

Select *File Open…* (for example: 2022 8 1 V1117 Her (Example)\\V1117Her_B.fit).

a. A dialog asks, “Do you want to estimate FWHM and Moffat β?” Select *Yes* to estimate these PSF parameters and automatically populate the corresponding Settings fields.
b. Image stretching uses *asinh* by default. Adjust the stretch as needed (screen display only; it does not modify the file).

You can optionally run *Photometry ->Iterative PSF Photometry* after loading an image, using the default PSF model (Circular Gaussian PRF or Moffat PSF).



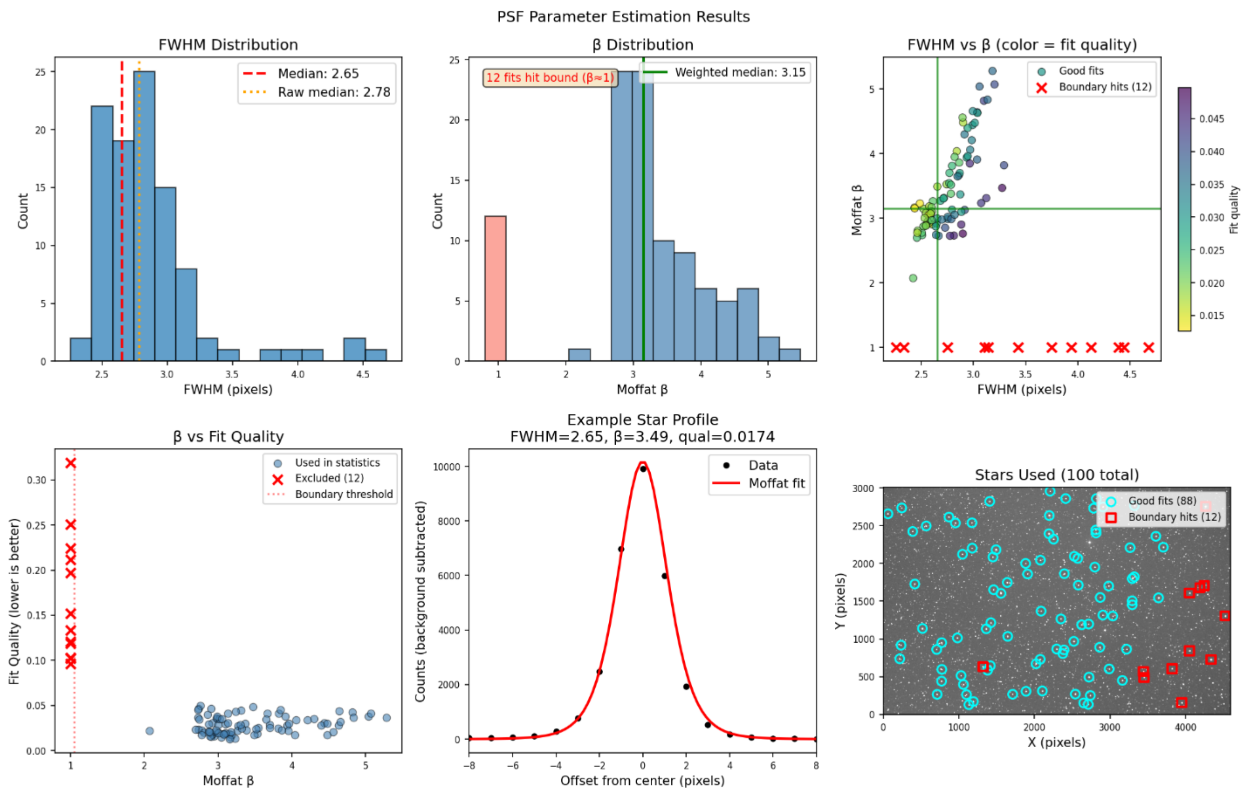
Use the mouse wheel to zoom in (wheel forward) or zoom out (wheel backward).

Center the image by Right Clicking the mouse.

To pan the image, hold *Shift* while dragging.

When a FITS file is opened and you select *Yes* in the prompt, MAOPhot runs PSF estimation and inserts the estimated FWHM and Moffat β values into Settings. A diagnostic plot showing the estimation statistics is also displayed.

PSF Parameter Estimation Diagnostic Plot (example)



This diagnostic plot summarizes the results of automatic PSF parameter estimation using Moffat-profile fitting across multiple stars in the image.

Panel Descriptions

FWHM Distribution (top-left): Histogram of measured Full Width at Half Maximum values in pixels. The red dashed line shows the final median FWHM used for photometry. An orange dotted line indicates the raw median before quality weighting, if different.

β Distribution (top-center): Histogram of Moffat β (beta) shape parameters. The green line shows the quality-weighted median. Red bars indicate boundary hits where the fit failed and β fell to its lower bound of 1.0; these are excluded from the final statistics.

FWHM vs. β (top-right): Scatter plot showing the relationship between FWHM and β for each star, with points colored by fit quality (yellow = better fit; purple = worse fit). Red X markers indicate boundary hits. Green crosshairs mark the final median values.

β vs. fit quality (bottom-left): Shows the relationship between β values and fit quality (lower values indicate better fits). Boundary hits ($\beta \le 1$) correspond to poor fits, supporting their exclusion from the summary statistics.

Example star profile (bottom-center): Cross-sectional profile through a representative well-fitted star, showing the observed data (black points) and the Moffat model (red curve). The title reports the FWHM, β, and fit quality for that star.

Stars used (bottom-right): Spatial distribution of stars used for PSF estimation overlaid on the image. Cyan circles indicate good fits included in the statistics; red squares mark boundary hits that were excluded.

5) [Optional] Find Peaks

Photometry-> Find Peaks (used when generating an effective PSF model)

a. MAOPhot uses DAOStarFinder to detect peaks, then removes peaks that exceed the Linearity Limit, are too close to the image edge, have close companions, or appear in a previously specified rejection list.

b. Example output: 186 peaks were found; 5 edge peaks were removed; and 10 peaks with close companions were removed. If *Max Number of Peaks* is set to 50, then the 50 brightest remaining peaks are used.

c. The remaining peaks are shown in the Selected Stars area. Review them and reject any peaks you do not want to use to build the ePSF model. To reject stars, click one or more entries and select Submit. To undo a rejection, click the entry again.

6) [Optional] Create Effective PSF

Photometry-> Create Effective PSF

a. Watch the progress bar in the output console (the console window opened when you launched MAOPhot).

- Reject stars that are not well isolated from their neighbors (many are rejected automatically).

- Optional: Save the rejection list (*Photometry->Save Rejection List…*).
- If needed, run *Photometry-> Create Effective PSF* again and repeat the review/rejection process.
- Inspect the Effective PSF Plot and confirm that the PSF shape looks reasonable.

Note: The ePSF model is built following Anderson and King (2000; PASP 112, 1360). The maximum number of iterations is hardcoded at 50. Any two peaks within an aperture width/height are rejected.

7) [Optional] Load Rejection List

Photometry -> Load Rejection List…

After loading, select *Photometry -> Create Effective PSF* again.

8) Perform Iterative PSF Photometry

If you did not create an effective PSF, the model defaults to either a Circular Gaussian PRF or a Moffat PSF. Select the corresponding radio button in Settings (*File-> Edit Settings…*).

Photometry -> Iterative PSF Photometry

a. Watch the progress bar in the output console.

b. When complete, MAOPhot saves the photometry results to a CSV (comma-separated values) file that you can inspect in Excel or another editor. In this example, the file is *V1117Her_B.fit.csv* (the *photometry file*).

About Iterative PSF Photometry: This uses the IterativePSFPhotometry class from Photutils, which is particularly useful for crowded fields. After the initial PSF fitted sources are subtracted from the image, new sources are detected in the residual image. The iterative process continues until no additional sources are detected or a maximum number of iterations is reached.

9) [Optional] Solve Image

Photometry->Solve Image (example files for Z Tau and V1117 Her already include WCS header data)

a. Run *Photometry->Solve Image* to plate-solve the image if the FITS file does not contain WCS header data.

b. Optional: After solving, use *File->Save* or *File->Save As…* to save the WCS header data in the FITS file.

Note: If the input image already includes WCS header data, you do not need to run *Solve Image*.

After solving, it is recommended that you save the image with the WCS data. Use *File->Save* or *File->Save As…* to save the FITS file.

10) Get Comparison Stars

Photometry->Get Comparison Stars (requires WCS header data)

a. Here, MAOPhot finds objects by querying AAVSO and VizieR servers.

b. The photometry file is updated with the objects found, associating the object, comp, and check stars with their photometry data.

> *For convenience, the qfit for each found object is displayed. This should be examined for high qfit values. "qfit" is a quality-of-fit metric defined as the sum of the absolute value of the fitted residuals divided by the fitted flux. qfit is zero for sources that are perfectly fit by the PSF model.*

c. Once this is done, the user can repeat steps 1 through 10 for another filter or immediately proceed to generate an AAVSO report for this single image.

11) Generate AAVSO Report

Generate AAVSO Report ->Single Image Photometry

a. Select the photometry CSV file generated in step 8 (e.g., *V1117Her_B.fit.csv*).

b. MAOPhot writes the AAVSO report to the *aavso_reports* subdirectory (e.g., 2022 8 1 V1117 Her (Example)\aavso_reports\AAVSO V1117Her_V V1117 Her_single.txt).

Example Single Image Photometry AAVSO Report:

#TYPE=Extended
#OBSCODE=ZZZZ
#SOFTWARE=Self-developed; MAOPhot 1.2.0 using photutils.psf
#DELIM=,
#DATE=JD
#OBSTYPE=CCD
#NAME,DATE,MAG,MERR,FILT,TRANS,MTYPE,CNAME,CMAG,KNAME,KMAG,AMASS,GROUP,CHART,NOTES
V1117 Her,2459793.599590,12.9885,0.0041,B,NO,STD,ENSEMBLE,na,120,13.0606,na,na,X27876MZ,Mittelman ATMoB Observatory|CMAGINS=-7.6458|CREFMAG=13.9688|KDEC=9.88044|KMAGINS=-8.554|KMAGSTD=13.0606|KRA=249.71804|KREFERR=0.009|KREFMAG=13.019|VMAGINS=-8.6261

Single Image Photometry does not utilize Transformation coefficients

Multi-Color Photometry Workflow

General workflow for multi-color photometry and AAVSO report generation, using the example “2022 8 1 V1117 Her (Example)”.

Execute steps 1 through 10 above for the B master (if not done already) and then the V master images. Skip step 11.

Then continue with step 12:

12) Perform Multi Color Photometry

Multi Color Photometry -> (B,V)

a. Select the 2 CSV files that were generated in step 10 (1 for B and 1 for V; e.g., V1117Her_B.fit.csv and V1117Her_V.fit.csv)

- When the file selection dialog appears, it will specify the filter needed

b. The output displays the "Multi Color Photometry" results

c. This data is saved in a "Master-Report": e.g., V1117 Her-BV-Master-Report.csv

d. Review the *outlier* column. If a comparison star is flagged as an outlier, it will be indicated there. MAOPhot uses the IQR (interquartile range) method to detect outliers.

e. If any outliers (comparison stars) were indicated, delete them from 'Select Comp Stars (AAVSO Label)' in the Settings if desired, then select 'Generate AAVSO Report->Multi Color Photometry->(B-V)' again.

13) Generate AAVSO Report

Generate AAVSO Report Multi Color Photometry -> (B,V)

a. Select the "Master-Report" CSV file that was generated in step 12 (e.g., V1117 Her-BV-Master-Report.csv)

- When the file selection dialog appears, it will specify the file needed using the default file names

b. An AAVSO report is generated in the subdirectory (e.g., 2022 8 1 V1117 Her (Example)/aavso_reports/AAVSO V1117Her_V V1117 Her.txt)

Example Color Photometry AAVSO Report:

```
#TYPE=Extended
#OBSCODE=ZZZZ
#SOFTWARE=Self-developed; MAOPhot 1.2.0 using photutils.psf
#DELIM=,
#DATE=JD
#OBSTYPE=CCD
#NAME,DATE,MAG,MERR,FILT,TRANS,MTYPE,CNAME,CMAG,KNAME,KMAG,AMASS,GROUP,CHART,NOTES
V1117 Her,2459793.599590,12.971,0.005,B,YES,STD,ENSEMBLE,na,120,13.067,na,na,X27876MZ,Mittelman ATMoB Observatory|KMAGINS=-8.554|KMAGSTD=13.067|KREFMAG=13.019|Tb_bv=0.039|Tb_bvErr=0.007|VMAGINS=-8.626
V1117 Her,2459793.589302,12.59,0.025,V,YES,STD,ENSEMBLE,na,120,12.072,na,na,X27876MZ,Mittelman ATMoB Observatory|KMAGINS=-9.407|KMAGSTD=12.072|KREFMAG=12.033|Tv_bv=-0.115|Tv_bvErr=0.008|VMAGINS=-8.96
```

More About Multi-Color Photometry

MAOPhot follows the approach used by VPhots TransformApplier (TA v2.71).

For example, for (B, V), MAOPhot solves the following two equations using GaussSeidel iteration:

Bs = bs + (Bc-bc) + Tb_bv * ((Bs-Vs)-(Bc-Vc))

Vs = vs + (Vc-vc) + Tv_bv * ((Bs-Vs)-(Bc-Vc))

In the equations above, s denotes the target star and c denotes the comparison star; uppercase letters are standard magnitudes and lowercase letters are instrumental magnitudes.

Note the TCs used are Tb_bv, and Tv_bv. These are in compliance with the AAVSO recommended transforms:

coeff	**B**	**V**	R	I
BVRI	Tb_bv	Tv_bv	Tr_vi	Ti_vi
BVR	Tb_bv	Tv_bv	Tvr	
BVI	Tb_bv	Tv_bv		Ti_vi
VRI		Tv_vi	Tr_vi	Tvi
BV	**Tb_bv**	**Tv_bv**		
VR		Tv_vr	Tr_vr	
VI		Tv_vi		Tvi

Error Estimation

For ensemble solutions with more than one comp star:

- Magnitude is estimated as the average of the individual comparison-star estimates.
- Error is estimated as the standard deviation of those estimates.

For one comp star:

- Error is estimated from the SNR of the target and the comparison star measurements.
- The standard error of a measurement is defined as 1/SNR.

Menu Functionality

File Menu

- **File->Open**: Load a FITS file into MAOPhot for analysis
- **File->Save**: Saves the loaded FITS file
- **File->Save As...**: Saves the loaded FITS file to a new file
- **File->Edit Settings...**: Opens the Settings window
- **File->Exit**: Exits MAOPhot

View Menu

- **View->Zoom In**: Zoom in by +0.5 scale increments (mouse wheel forward)

- **View->Zoom Out**: Zoom out by 0.5 scale increments (mouse wheel backward)
- **View->100% Zoom**: Zoom to normal scale
- **Invert**: Inverts the loaded image
- **View->Refresh**: Reloads the photometry file (if it exists)

Photometry Menu

- **Photometry->Find Peaks**: Uses DAOStarFinder to search for peaks in the image that will be used for effective PSF (ePSF) model generation. It discards any peaks over the Linearity Limit, any that are close to the edge, and any that have close companions. It also discards any that the user has specified previously.
- **Effective PSF** > **Create**: Analyzes the image and generates an ePSF model following Anderson and King (2000; PASP 112, 1360). The maximum number of iterations is hardcoded at 50. Any two peaks within one aperture width/height are rejected. If a rejection list has been loaded, those peaks are also excluded.
- **Effective PSF** > **Load...**: Loads a saved ePSF file
- **Effective PSF** > **Save As...**: Saves the loaded ePSF to a file
- **Effective PSF** > **Clear**: Clears the PSF model, rejection lists, and plot
- **Photometry->Load Rejection List...**: Loads a previously saved rejection list
- **Photometry->Save Rejection List...**: After running "Photometry->Find Peaks", the user can select peaks to be rejected by mouse-clicking on them in the "Selected Stars" area. When user clicks on a peak to be rejected, the word "Rejected" appears over that selection. The user can page through the list of peaks and reject (and undo a rejection). When the Submit button is clicked, the peaks rejected will not be used in the next "Photometry->Create Effective PSF". Once the rejected peaks are submitted, they can be saved using "Photometry->Save Rejection List..."
- **Photometry->Iterative PSF Photometry**: Executes the iterative version of PSF Photometry using Photutils' IterativePSFPhotometry class, where new sources are detected in the residual image after the fitted sources are subtracted. This is essential for crowded field photometry.
- **Photometry->Solve Image**: Uses Astrometry.net server to add WCS Header information
- **Photometry->Get Comparison Stars**: Queries AAVSO and VizieR for VSX objects and comparison stars in the field

Multi Color Photometry Menu

- **Multi Color Photometry->(B,V)**: Executes Multi Color photometry for B and V
- **Multi Color Photometry->(V,R)**: Executes Multi Color photometry for V and R
- **Multi Color Photometry->(V,I)**: Executes Multi Color photometry for V and I
- **Multi Color Photometry->(B,V,R)**: Executes Multi Color photometry for B, V, and R
- **Multi Color Photometry->(B,V,I)**: Executes Multi Color photometry for B, V, and I
- **Multi Color Photometry->(V,R,I)**: Executes Multi Color photometry for V, R, and I
- **Multi Color Photometry->(B,V,R,I)**: Executes Multi Color photometry for B, V, R, and I

Generate AAVSO Report Menu

- **Generate AAVSO Report->Single Image Photometry**: Generates an AAVSO report in extended format for a single filter
- **Generate AAVSO Report->Multi Color Photometry->(B,V)**: Generates an AAVSO report for B and V
- **Generate AAVSO Report->Multi Color Photometry->(V,R)**: Generates an AAVSO report for V and R
- **Generate AAVSO Report->Multi Color Photometry->(V,I)**: Generates an AAVSO report for V and I
- **Generate AAVSO Report->Multi Color Photometry->(B,V,R)**: Generates an AAVSO report for B, V, and R
- **Generate AAVSO Report->Multi Color Photometry->(B,V,I)**: Generates an AAVSO report for B, V, and I
- **Generate AAVSO Report->Multi Color Photometry->(V,R,I)**: Generates an AAVSO report for V, R, and I
- **Generate AAVSO Report->Multi Color Photometry->(B,V,R,I)**: Generates an AAVSO report for B, V, R, and I

List of Parameters in Settings Window

Parameter		Description	Units	Required*
Max Number of Peaks		Maximum number of peaks displayed in the Selected Stars area after you run *Photometry > Find Peaks*.	Integer	
Fitting Width/Height		Width and height of the rectangular region around a stars centroid used for PSF fitting.	pixels	
Maximum Ensemble Magnitude		Magnitude limit used when fetching comparison stars.	Magnitude	
FWHM		PSF photometry searches for peaks with similar FWHM. This value is used by DAOStarFinder to define the Gaussian kernel.	pixels	
Oversampling		Integer oversampling factor of the ePSF model relative to the input stars along each axis (see Technical Notes).	Integer	
DAOStarFinder Threshold Factor		Absolute image value above which sources are selected, expressed as a multiple of the image standard deviation.	Float	
Photometry Iterations		Number of iterations performed by Iterative PSF Photometry.	Integer	
Lower Bound for Sharpness		Lower bound on sharpness for DAOStarFinder detection (upper bound is fixed at 1.0). Used only with IterativePSFPhotometry.	Float	
Matching Radius		Maximum allowed separation between an image coordinate and a catalog coordinate for a match.	arcsec	
PSF Fitter		Fitter algorithm used by Iterative PSF Photometry:	List selection	

Parameter		Description	Units	Required*
		TRF LS: Trust Region Reflective algorithm with a least-squares statistic (common default) **Sequential LS Programming**: SLSQP optimization with a least-squares statistic (useful in crowded fields) **Simplex LS**: Simplex optimization with a least-squares statistic (useful for noisy images)		
Max qfit		Discards any PSF fit with qfit greater than the specified maximum. qfit is a quality-of-fit metric defined as the sum of the absolute value of the fit residuals divided by the fitted flux. A perfectly fit source has qfit = 0.	Float	
Min Separation Bias		Additional bias term added to the critical separation distance used for grouping sources for simultaneous fitting. Critical separation is computed as (FWHM/2 + Fitting Width/2 + Min Separation Bias). Sources separated by less than this value (in pixels) are placed in the same group and fitted together. Increase this value if overlapping sources are not being grouped (which can lead to poor fits and high qfit values).	Float	
Default PSF Model		Selects the default analytic model used when an effective PSF (ePSF) is not loaded or has not been created.	Radio buttons	
Circular Gaussian PRF		This model is evaluated by integrating the 2D Gaussian over the input coordinate pixels and is equivalent to assuming the PSF is 2D Gaussian at a *sub-pixel* level. Because it is	Radio button	

Parameter		Description	Units	Required*
		integrated over pixels, this model is considered a PRF instead of a PSF.		
Moffat PSF		This model is evaluated by sampling the 2D Moffat function at the input coordinates. The Moffat profile is normalized such that the analytical integral over the entire 2D plane is equal to the total flux.	Radio button	
beta		Moffat β (beta): controls the slope of the profile wings at large radii. Larger values place less flux in the wings; for large β the profile approaches a Gaussian. β must be greater than 1 for a finite integral. (β = 1 corresponds to a Lorentzian, whose integral diverges.)	Float	
Generate Residual Image		If selected, saves a residual image in the working directory after IterativePSFPhotometry completes. The residual image shows the remaining signal after subtracting the fitted PSF models and can be used to assess fit quality.	Checkbox	
Telescope		Telescope name (reference only; not used in calculations).	String	
Tb_bv		Transformation Coefficient	float	Yes
Tb_br		Transformation Coefficient	float	Yes
Tb_bi		Transformation Coefficient	float	Yes
Tv_bv		Transformation Coefficient	float	Yes
Tv_vr		Transformation Coefficient	float	Yes

Parameter		Description	Units	Required*
Tr_vr		Transformation Coefficient	float	Yes
Tr_ri		Transformation Coefficient	float	Yes
Ti_ri		Transformation Coefficient	float	Yes
Tv_vi		Transformation Coefficient	float	Yes
Ti_vi		Transformation Coefficient	float	Yes
Tr_vi		Transformation Coefficient	float	Yes
Tbv		Transformation Coefficient	float	Yes
Tbr		Transformation Coefficient	float	Yes
Tbi		Transformation Coefficient	float	Yes
Tvr		Transformation Coefficient	float	Yes
Tri		Transformation Coefficient	float	Yes
Tvi		Transformation Coefficient	float	Yes
Extinction Coefficients		Extinction coefficients for filters B, V, R, I, and C (currently not used).		
AAVSO Observer Code		AAVSO observer code written to the report as #OBSCODE.	String	Yes
Exposure Time		Exposure time, usually read from the FITS header. Used to compute instrumental magnitude.	Float	Yes
Airmass		Airmass value, typically read from the FITS header. Written as “na” in the report if unavailable.	Float	

Parameter		Description	Units	Required*
Date-Obs (JD)		Observation time in Julian Date (JD), read from the FITS header (read-only).	JD	Yes
Date-Obs (UTC)		Observation time in UTC, read from the FITS header (read-only).	UTC	
Notes		Notes written to the report NOTES field.	String	Yes
Comparison Catalog		**AAVSO**: Use comparison stars from the AAVSO Variable Star Database. **Gaia DR2**: Use objects from VizieR catalog I/345 (Gaia DR2; Gaia Collaboration 2018). **APASS DR9**: Use objects from VizieR catalog II/336 (APASS DR9; Henden+ 2016). Note: R and I are Sloan; comparison names include a “B” prefix (e.g., 113B_2). **APASS DR10**: Use APASS DR10 objects. Note: R and I are Sloan; comparison names include an “A” prefix (e.g., 113A_2).	List selection	
AAVSO ChartID		Optional AAVSO chart ID to use when retrieving comparison-star data (e.g., X28484CPQ).	String	
From FITS		If selected, reads CCD Filter from the FITS header keyword FILTER. If cleared, you can manually enter the CCD Filter value.	Checkbox	
CCD Filter		Filter used for the image (usually read from the FITS header).	String	Yes
Object Name		Name of the target (variable) star to measure.	String	Yes
α		Right ascension (RA) of the target object. If the object name is not found in the VSX catalog, MAOPhot uses α and δ for the search.	hmsdms	

Parameter		Description	Units	Required*
δ		Declination (Dec) of the target object. If the object name is not found in the VSX catalog, MAOPhot uses α and δ for the search.	hmsdms	
Use Check Star		AAVSO label of the check star (written to the report as KNAME).	AAVSO label	Yes
Select Comp Stars		Comma-separated list of AAVSO labels for comparison stars to use. If more than one is provided, the report uses the ENSEMBLE measurement type.	AAVSO labels	Yes
Minimum SNR		Minimum SNR required for comparison and check stars. Stars with SNR below this value are excluded.	Float	
Display selected objects only		Displays only the objects specified by Object Name, Select Comp Stars, and Use Check Star.	Radio button	
Display all objects		Displays all objects found by plate solving. Duplicate names may be suffixed with .0, .1, etc.	Radio button	
Astrometry.net Server		Astrometry.net server host name (e.g., nova.astrometry.net) or URL for a local server.	String	
Astrometry.net API Key		API key for astroquery.astrometry.net. Create an account at astrometry.net, copy your key from your profile, and paste it here.	String	
Settings File Name		Name of the most recently loaded settings file (informational only). To load settings, use the *Load…* button.	String	
User Note:		Optional note for your reference (not used in calculations).	String	

*Required: These settings are inserted into the AAVSO report; most are automatically filled from the FITS header. (Only some transformation coefficients are inserted into the AAVSO report.)

Keyboard Shortcuts

View: Zoom In	Mouse wheel forward
View: Zoom Out	Mouse wheel backward
Center Image	Right Button Click
Drag Image	Shift + drag

Definitions

Term	Definition
IRAF	IRAF (Image Reduction and Analysis Facility) provides a wide range of astronomical image-processing tools. It was developed for the astronomical community, and researchers in other scientific fields have also used it for general image processing. IRAF Community: https://iraf-community.github.io/

Technical Notes

MAOPhot source code: https://github.com/petefleurant/PSF-Photometry

Optimization Algorithms: gtol and xtol

In optimization algorithms used in statistics and numerical computing, **gtol (gradient tolerance)** and **xtol (step tolerance)** are termination conditions that determine when an algorithm should stop iterating.

1. gtol (Gradient Tolerance)

- **Definition**: The optimization stops when the gradient (first derivative) of the objective function is small.
- **Interpretation**: The gradient represents the rate of change of the function. A small gradient means the function has reached a local minimum (or is very close to it).
- **Usage**: If $||\nabla f(x)|| < gtol$, the algorithm stops.
- **Common in**: Newton methods, Quasi-Newton methods (e.g., BFGS, L-BFGS), and Conjugate Gradient.
- **Example**: If gtol = 1e-6, the optimizer stops when the gradient norm is smaller than 0.000001.

2. xtol (Step Tolerance)

- **Definition**: The optimization stops when the change in the parameter values (x) is very small.
- **Interpretation**: If the steps taken in each iteration become very small, the algorithm assumes convergence.
- **Usage**: If $||x_{new} - x_{old}|| < xtol$, the algorithm stops.
- **Common in**: Trust-region and Line Search methods.

- **Example**: If xtol = 1e-8, the optimizer stops when the parameter updates are smaller than 0.00000001.

Key Differences

Termination Condition	Meaning	Stops when...	Typical Use
gtol	Gradient tolerance	The gradient is small (function slope is near 0)	Gradient-based methods (e.g., BFGS, CG)
xtol	Step tolerance	The parameter change is small	Trust-region & Line Search

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Bradley, L., et al. (2023) "Photutils: Photometry tools" Astrophysics Source Code Library https://photutils.readthedocs.io/

Astropy Collaboration (2022) "The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package" The Astrophysical Journal, 935, 167 DOI: 10.3847/1538-4357/ac7c74

Additional Resources

Documentation Links

- **Photutils PSF Photometry Guide**: https://photutils.readthedocs.io/en/stable/user_guide/psf.html
- **Astropy Modeling and Fitting**: https://docs.astropy.org/en/stable/modeling/
- **AAVSO Extended Format**: https://www.aavso.org/aavso-extended-file-format
- **AAVSO WebObs**: https://www.aavso.org/webobs

- **Astrometry.net**: https://nova.astrometry.net/

Oversampling

Oversampling controls how finely the ePSF model is sampled relative to your image pixels. It is set under File → Edit Settings and defaults to 2.

When oversampling is set to 2, each image pixel is represented by a 2×2 grid in the ePSF model, giving the builder more resolution to capture the true PSF shape. Higher values (3, 4, ...) create even finer grids but increase computation time and memory use with diminishing returns.

For most users, the default value of 2 is appropriate. At a typical image scale of ~0.8"/pixel with seeing-limited FWHM of 3–4 pixels, stellar profiles are well sampled — the PSF is spread across enough pixels for the ePSF builder to reconstruct its shape reliably. Increasing oversampling beyond 2 in this regime provides little practical benefit.

You would consider increasing oversampling to 3 or higher only if your stars are significantly undersampled — for example, if your FWHM is below about 2 pixels, where the PSF shape is not adequately captured at the native pixel scale. This situation can arise with very short focal lengths or exceptionally good seeing on a coarse detector.

If your FWHM is 3 pixels or more, the default oversampling value of 2 is typically appropriate.

Tip: If the ePSF Normalization Index is consistently much greater than 1.00 or much less than 1.00, try adjusting the oversampling value.

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